Module 1: Accounts and Programs

Core Concepts in Solana

Accounts vs Programs

Solana's Account Model

- **Accounts**: Data storage containers (like database records)
- **Programs**: Executable code (like smart contracts)
- **Key Difference**: Accounts can hold both data AND code

EVM Comparison

- **EVM**: Contracts hold code, separate storage
- **SVM**: Accounts are unified - they can be programs OR data

Account Types

1. User Accounts (External Wallets)

- Controlled by private keys
- Hold SOL and tokens
- Can sign transactions
- Created by System Program

2. Program Accounts

- Contain executable bytecode
- Immutable once deployed
- Executed by Solana runtime
- Cannot sign transactions directly

Account Types (continued)

3. Program Derived Addresses (PDAs)

- Deterministically derived from seeds + program ID
- No private key exists
- Controlled by programs
- Enable program-controlled accounts

4. System Accounts

- Built-in Solana programs
- System Program, SPL Token Program, etc.
- Handle core blockchain operations

Account States

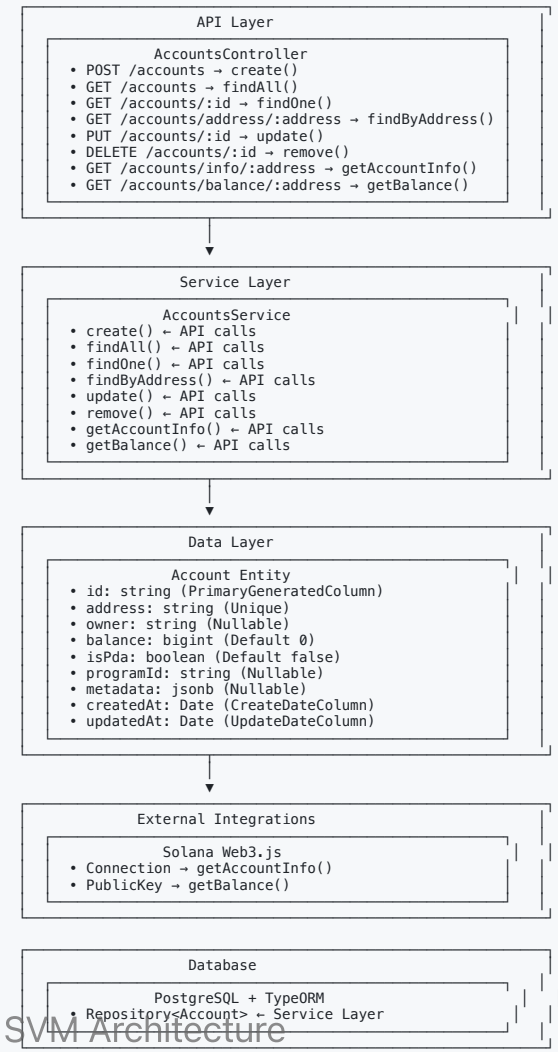
Account Lifecycle

- **Uninitialized**: No data, zero balance
- **Active**: Contains data, has balance
- **Frozen**: Temporarily locked
- **Closed**: Zero balance, can be garbage collected

Rent Exemption

- Accounts must maintain minimum balance
- Prevents storage spam
- "Rent" paid to validators for storage

Architecture Overview



API Endpoints

Account Management

- `POST /accounts` - Create new account record
- `GET /accounts` - List all accounts
- `GET /accounts/:id` - Get account by ID
- `GET /accounts/address/:address` - Get account by Solana address
- `PUT /accounts/:id` - Update account metadata
- `DELETE /accounts/:id` - Remove account record

Blockchain Queries

- `GET /accounts/info/:address` - Get on-chain account info
- `GET /accounts/balance/:address` - Get SOL balance

Database Schema

Account Entity Fields

- `id` : Primary key (UUID)
- `address` : Solana public key (unique)
- `owner` : Account owner address
- `balance` : SOL balance (bigint)
- `isPda` : Program Derived Address flag
- `programId` : Associated program ID
- `metadata` : Additional JSON data
- `createdAt/updatedAt` : Timestamps

Key Implementation Concepts

PDA Generation

```
// Derive PDA from seeds and program ID
const [pdaAddress, bump] = PublicKey.findProgramAddressSync(
  [Buffer.from("escrow"), userPublicKey.toBuffer()],
  programId
);
```

Account Info Retrieval

```
// Get account information from blockchain
const accountInfo = await connection.getAccountInfo(publicKey);
if (accountInfo) {
  // Account exists with data
  console.log('Balance:', accountInfo.lamports);
  console.log('Owner:', accountInfo.owner.toString());
}
```

Common Patterns

Account Creation Flow

1. Generate keypair or derive PDA
2. Check if account exists
3. Create account record in database
4. Return account information

Balance Checking

1. Query Solana RPC for account info
2. Convert lamports to SOL
3. Cache result for performance
4. Return formatted balance

Security Considerations

Access Control

- Validate account ownership
- Implement proper authorization
- Rate limit API calls

Data Validation

- Verify Solana addresses
- Sanitize metadata input
- Prevent SQL injection

Next Steps

Module 1 Implementation Tasks

- ✅ Basic account CRUD operations
- ✅ Solana RPC integration
- ✅ PDA generation service
- ✅ Account abstraction features

Related Modules

- **Module 2**: Transactions & Instructions
- **Module 3**: Token Standards
- **Module 4**: Account Abstraction

Thank You!

Questions?

Ready to explore Transactions & Instructions?

Next: Module 2 - Transactions & Instructions ➡️